AI-Based Hyper-Local Aviation Weather Forecast Verification System for SL M

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EXECUTIVE ABSTRACT & SYSTEM SCOPE

This research presents an AI-based hyper-local weather forecasting and automated verification system tailored for Bandaranaike International Airport (VCBI), Sri Lanka. Historical meteorological observations and operational METAR/TAF records are utilized to engineer parameter-specific machine learning models. Predictions are rigorously validated and exposed via an operational verification dashboard.

1. Introduction and Operational Challenge

Aviation weather forecasts in tropical equatorial zones like Sri Lanka encounter abrupt monsoonal shifts, localized squalls, and severe wind shears. Conventional global weather models lack the spatial resolution required for pinpoint aerodrome operations.

Project R26-IT-132 closes this gap by implementing an integrated multi-model architecture capable of continuously comparing terminal area predictions against actual METAR ground-truth records.

2. Multi-Model Architecture & Individual Contributions

" IT22113986: Coupled Temperature and Pressure prediction using Hybrid Random Forest - LSTM achieving R2 of 0.84 and MAE of 0.86, translating directly to density altitude calculations.

" IT22269720: Wind vector and runway component prediction using XGBoost and Temporal Fusion Transformers (TFT), achieving MAE under 1.0 knot for runway 04/22 crosswind safety.

" IT22619976: Cloud ceiling and rapid visibility change classification using tree ensembles, reaching 78.15% classification accuracy for low-visibility approach warnings.

" IT22069504: Humidity, QNH altimeter setting and dew-point modeling via LightGBM (MAE 0.58 deg C for dew point, 1.8 hPa for QNH) with physical relative humidity derivation.

3. Empirical Verification Results & Conclusion

The proposed system demonstrably outperforms standard baseline numerical models across all critical flight parameters. The verification engine provides quantitative skill scores (MAE, RMSE, R2, Threat Score), offering air traffic managers and pilots real-time operational situational awareness.